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Additive effects of depression and abdominal obesity on cognitive function in middle-aged and older population: evidence from multinational cohorts

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Abstract

Background This study aimed to investigate the joint trajectories of obesity/abdominal obesity and depression, and their association with cognitive function among four nationally representative cohorts.

Methods We used data from four nationally representative cohorts (China, UK, USA, and Mexico) in adults over the age of 45, which included a total of 114,633 participants. Kml3D clustering algorithm was conducted to identify the potential joint trajectories of obesity/abdominal obesity and depression of homogeneous groups. Generalized Estimating Equations (GEE) were performed to examine the joint trajectory of obesity/abdominal obesity and depression in relation to cognitive function.

Results In all cohorts, the baseline “Comorbidity” (with both depression and obesity/abdominal obesity) exhibited significantly poorer performance on subsequent cognitive assessments compared to the “Neither condition” group (neither depression nor obesity). Cluster analysis and GEE revealed that when Body Mass Index (BMI) was used as an obesity indicator, individuals in the joint trajectory groups with depression trajectories (regardless of obesity trajectories) across four cohorts exhibited poorer cognitive performance compared to the Normal weight and No depressed group (CHARLS: $\beta = -0.35$, 95% CI -0.42 to -0.28 ; ELSA: $\beta = -0.32$, 95% CI -0.44 to -0.20 ; HRS: $\beta = -0.20$, 95% CI -0.27 to -0.13 , MHAS: $\beta = -0.15$, 95% CI -0.19 to -0.10 ; all $P < 0.001$). Conversely, associations between the joint trajectory groups with obesity trajectories (regardless of depression trajectories) and cognitive function demonstrated significant heterogeneity across cohorts. Abdominal obesity measures indicated that the abdominal obesity or higher waist-to-height ratio (WHtR) and Depression group significantly contributed to cognitive decline compared to those in the No abdominal obesity and No depression group (CHARLS: $\beta = -0.31$, 95% CI -0.39 to -0.23 ; ELSA: $\beta = -0.20$, 95% CI -0.29 to -0.12 ; HRS: $\beta = -0.20$, 95% CI -0.27 to -0.14 , all $P < 0.001$).

Conclusions Abdominal obesity and depression exert independent additive and fluctuating effects on measures of cognition in middle-aged and older persons. Strategies that broadly aim to decrease excess fat, notably abdominal obesity, represent near-term interventions that may beneficially influence aspects of cognition in depression.

Keywords Trajectory, Obesity, Abdominal obesity, Depression, Cognitive function, Cohort study

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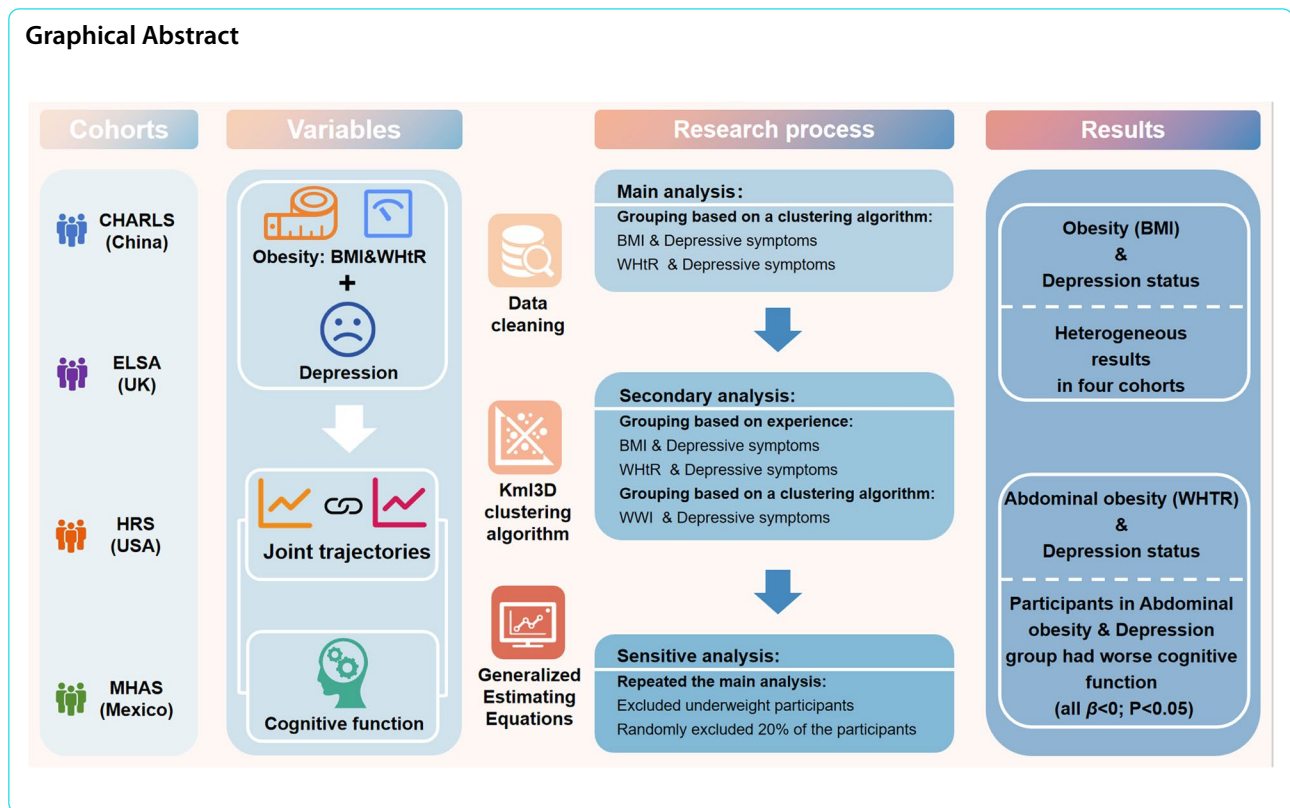
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Graphical Abstract



Background

The widespread nature of depression and obesity carries considerable public health challenges [1, 2], particularly among middle-aged and older adults [3]. Recent US data show that 44.3% of adults aged 40–59 were classified as obese during 2017–2020, increasing to nearly 41.5% among those aged 60 and older [4]. Similarly, a recent study found that 21.3% of adults aged 50 and above report depressive symptoms [5].

Furthermore, both disorders co-occur within individuals, with depression conferring a 37% increased risk of obesity and obesity elevating depression risk by 40% [1], creating compound impairment in physical and mental health. This interplay is mediated mainly through shared genetic loci [6], neuroendocrine dysfunction [notably the hypothalamic–pituitary–adrenal (HPA) axis hyperactivity], metabolic disturbances, and chronic inflammation [7]. These interconnected pathways form a self-perpetuating vicious cycle that exacerbates pathological progression in both conditions [8]. Accumulating evidence suggested that the comorbidity of obesity and depression may impair health outcomes in middle-aged and older adults, including increased risks of functional disability, cardiometabolic multimorbidity, and cognitive decline [9–12]. These interconnected health burdens underscore the urgent need

for integrated preventive and therapeutic strategies targeting this vulnerable demographic.

Cognitive decline is a growing public-health concern. Statistics show that approximately 5–8% of older adults experience cognitive decline annually, which is considered a precursor symptom of dementia [13]. Neurodegenerative processes mediated by chronic inflammation and metabolic disturbances, including cerebrovascular pathology, are key contributors to this decline [14–16]. Given the established mechanistic associations of obesity and depression with chronic inflammation and metabolic dysfunction [17], these conditions may synergistically drive cognitive impairment progression.

Extensive evidence indicates that depression links to cognitive decline and is a prodrome to dementia [18–20]. Depressive symptoms mediate cognitive impairment via multiple pathways, including the aforementioned HPA axis hyperactivation, inflammation, and hippocampal atrophy [21–23]. Chronic or recurrent depression particularly impairs attention, executive function, and episodic memory—specific domains critical for daily functioning and predictive of dementia risk [24, 25]. However, the association between obesity and cognitive health in middle-aged and older adults remains unclear [9]. Some studies suggest that obesity may have a protective effect on cognitive decline [26, 27], supporting the

concept of an “obesity paradox” [28], while others view obesity as a risk factor for cognitive decline [27, 29, 30], necessitating further investigation into their connection. Of note, while Body Mass Index (BMI) is a widely accepted indicator for defining obesity, cross-cultural researches have uncovered heterogeneity in its correlation with cognitive deterioration [9, 31]. Importantly, as a contributing factor to various chronic illnesses including type 2 diabetes, cardiovascular disease, and metabolic syndrome [32], abdominal obesity (AO) [33] has been demonstrated by numerous studies to have a robust association with cognitive decline [34–37]. Additionally, considering the comorbid mechanisms between obesity and depression, the joint changes of both conditions over time should not be overlooked.

To overcome limitations of prior studies using a single obesity indicator and elucidate mechanisms underlying cognitive decline, this study incorporates BMI and (waist to height ratio) WHtR as predictive indicators in middle-aged and older populations. While prior work often has predominantly examined the effects of a single developmental trajectory of obesity or depression on cognitive decline, it overlooked the combined effects of their joint trajectories. Hence, we conducted a multinational cohort study to examine the association between comorbid obesity-depression trajectories and cognitive decline. We hypothesize that individuals exhibiting co-occurring trajectories of persistent obesity/AO and persistent depression demonstrate significantly poorer cognitive performance compared to those who remained consistently non-obese and non-depressed throughout the observation period. Findings aim to identify high-risk populations and inform targeted interventions for age-related cognitive impairment.

Methods

Study population

The multinational cohort employed in this survey design comprised four prospective, nationally representative longitudinal studies: the China Health and Retirement Longitudinal Study (CHARLS, China), the English Longitudinal Study of Ageing (ELSA, United Kingdom), the Health and Retirement Study (HRS, United States), and the Mexican Health and Aging Study (MHAS, Mexico). All sample population was recruited as part of a stratified, multistage probability design, with longitudinal follow-up administered by trained research teams. The detailed introduction of each cohort could be found on their websites [CHARLS: <https://charls.pku.edu.cn/gy/gyxm.htm> [38]; ELSA: <https://ukdataservice.ac.uk/about/> [39]; HRS: <https://hrs.isr.umich.edu/about/> [40]; MHAS: <https://www.mhasweb.org/Home/Index.aspx> [41]]. Relevant ethical review institutes granted our survey's ethical

approval, and all participants signed informed consent forms before the survey.

According to the current research objectives, we ruled out the impact of COVID-19 and selected the appropriate waves from the four cohorts. The included waves need to include all the key variables of this study (e.g., BMI, WHtR, depressive symptoms, cognitive function scores, and covariates). Thus, 3 waves (wave1–3: 2011, 2013, and 2015) in CHARLS, 3 waves (wave3–5: 2012, 2015, and 2018) in MHAS, 4 waves (wave2,4,6,8: 2004, 2008, 2012, and 2016) in ELSA, and 6 waves (wave8–13: 2006–2016) in HRS (participants only received body measurements in odd or even waves, so we merged the odd and even waves into three waves: wave1, wave2, wave3: 2006/2008, 2010/2012 and 2014/2016) were included in the current study. Our exclusion criteria were as follows: (1) age < 45 years old; (2) participants with cognitive-related disorders (see Additional file 1: Table S1) [42]; (3) cases of missing, unclear responses, and rejections both in baseline and follow-up surveys. Therefore, 114,633 participants from four countries were considered, and after screening, a total of 36,718 participants remained. The comprehensive screening flowchart of participants could be seen in Additional file 1: Fig. S1, with an in-depth description of each cohort's selecting procedure available in Additional file 1: Fig. S2.

Obesity and abdominal obesity

In the main analyses, obesity in middle-aged and older adults was defined by Body Mass Index (BMI), and the calculation of BMI was done by dividing a person's weight (in kilograms) by the square of their height (in meters). It is worth noting that objectively measured height, weight, and waist circumference (WC) were provided only in CHARLS, ELSA, and HRS. MHAS only included self-reported height and weight (no WC data available). Furthermore, because the study population comes from various continents and exhibits variations in ethnicity, physique, culture, and public health priorities, it is considered to adopt distinct body type classification standards [43]. Our study referred to previous research, stipulated the use of the Chinese criteria to classify participants from CHARLS, and obesity is defined as $BMI \geq 28 \text{ kg/cm}^2$ [44]. For the population from ELSA, HRS, and MHAS cohorts, the European and American criteria were utilized for classification, and obesity is defined as $BMI \geq 30 \text{ kg/cm}^2$ [45] (cut-off values of BMI under different standards were shown in Additional file 1: Table S2) [44].

Abdominal obesity was defined by waist-to-height ratio (WHtR), which was calculated as WC divided by height (both in centimeters). Considering it is probable that middle-aged and older adults may have an overestimation

of abdominal obesity due to height loss [46]. The current study defined 0.6 as the cut-off value of abdominal obesity [47].

In addition, we opted to use the Weight-Adjusted Waist Index (WWI) in the secondary analysis to reflect obesity [48]. The WWI standardizes WC with weight, providing a more accurate depiction of adiposity that is unrelated to weight and partially circumvents the “obesity paradox” associated with BMI [49]. It was calculated by dividing WC (in centimeters) by the square root of weight (in kilograms) ($\sqrt{\text{kg}}$) [50].

Depressive symptoms assessment

In all four cohorts, depressive symptoms and depression were evaluated by calculating the total scores of different standardized scales and referring to the corresponding cut-off values. Briefly, in CHARLS, depressive symptoms were measured by the ten-item Center for Epidemiologic Studies Depression Scale (CESD-10); in ELSA and HRS, depressive symptoms were measured by the eight-item version of the Center for Epidemiologic Studies Depression Scale (CES-D); in MHAS, depressive symptoms were measured by the nine-item scale modified from the Center for Epidemiological Studies Depression scale (CESD-9). Additional file 1: Table S3 [20, 51–55] provides more information about each scale.

Empirical classification (secondary analysis): depression and obesity group

In the baseline and outcome wave, participants’ depression and obesity status were divided into four groups based on previous research and empirical criteria: (1) neither depression nor obesity (“Neither condition”); (2) depression without obesity (“Depression alone”); (3) obesity without depression (“Obesity alone”); (4) with both depression and obesity (“Comorbidity”) (3). Therefore, the changes in the subjects’ depression-obesity status can be further categorized into three groups over time: (1) Stability; (2) Progression; (3) Regression. More details about the grouping criteria are provided in the Additionalfile 1: Fig. S3.

Cognitive function assessment

Cognitive function was assessed through face-to-face self-questionnaire, evaluating the response and operational performance of participants. Each cohort used different questionnaires and measurement dimensions. Briefly, in CHARLS, the cognitive function was assessed by the Chinese version of the Mini-Mental State Examination (MMSE); in ELSA, the cognitive function assessment was assessed by standardized episodic memory and orientation tests; in HRS, the cognitive function was assessed by the modified version of the Telephone

Interview for Cognitive Status (TICS); in MHAS, the cognitive function was assessed by the Cross-Cultural Cognitive Examination (CCCE). In all tests, a higher cognitive score indicated better cognitive performance. To achieve comparability across different tests, we generated the z-score of cognitive function for each cohort, referring to previous research [20]. Details of cognitive assessments for each cohort and the method for calculating z-score are provided in the Additional file 1: Table S4 and Supplementary Methods1 [20, 42, 56–61].

Covariates

Covariate selection was guided by two criteria: (1) availability of key demographic variables in the database, and (2) prior empirical evidence from relevant studies [42, 62]. The following variables were ultimately incorporated as potential confounding factors: age, gender, race, marital status, education level, residential area, household financial situation, five doctor-diagnosed health problems (hypertension, diabetes, cancer, stroke and arthritis), smoking and drinking status at baseline. The definition of covariates is reported in the Additionalfile 1: Supplementary Methods2.

Statistical analysis

All analyses were conducted based on complete datasets with exclusion of missing data. In the main analyses, the baseline characteristics of the participants were described as means $\pm$ standard deviations (SD) of continuous variables and percentages (%) of categorical variables. Differences in the characteristics were analyzed using t-tests and chi-square tests. And to better present the transition of depression-obesity status, we depicted Sankey diagrams to visualize the proportion of participants transitioning across different groups from baseline to outcome wave.

To identify the potential joint variable-trajectories of obesity (indicator: BMI and WHtR) and depression of homogeneous groups, the current study used Kml3D, a partitioning clustering algorithm from the package Kml3D [63], which is based on the k-means algorithm [64]. Then, we thoroughly considered both Calinski & Harabasz criterion (high value denoting good partition) [65] and the interpretability of the results to decide the final number of clusters for each cohort; meanwhile, we named the trajectory groups for further analysis. Further explanations on this methodology are provided in the Additionalfile 1: Supplementary Methods3 [66, 67].

Subsequently, Generalized Estimating Equations (GEE), which are an extension of generalized linear models (GLM) and allow for the resolution of the issue of the correlations between observations in longitudinal studies [68], were employed to evaluate baseline

depression-obesity status of participants and the joint trajectory of the two in relation to cognitive function (z-score). We set follow-up time as the timescale to calculate the β with corresponding 95% confidence intervals (CI). GEE was repeated after stratifying by baseline main characteristics.

In subgroup analyses, we stratified populations based on main demographic variables (e.g., age, sex, educational attainment) to examine heterogeneous associations between joint trajectories of BMI/WHtR and depressive symptoms and cognitive function change across distinct population subgroups. The interaction effects between different subgroups and combined trajectory groups were examined. In secondary analyses, we employed various obesity indicators and different grouping strategies to redefine changes in depression-obesity status, aiming to supplement the main results. (1) Considering that KmL3D is a data-driven clustering algorithm and might ignore practical experience standards, we classified participants into stability groups, progression groups, and regression groups based on a previous classification method and empirical criteria, reflecting the transition between obesity and depression at baseline and outcome waves. We then repeated GEE to explore the association between the transition groups and development of cognitive function in participants. (2) Used WWI as another obesity indicator and generated joint trajectory groups. Subsequently, replicated the statistics of the main results. We also conducted several sensitivity analyses to test the robustness of our findings: (1) Removed the underweight population (BMI < 18.5) and the statistical approach was consistent with previous work; (2) To ensure joint trajectories were robust, 20% of the participants were randomly excluded and repeated the KmL3D cluster analysis [67].

All analyses were conducted with R (version 4.3.1); specifically, we used “kml3d”, “gee”, “geepack” and “ggplot2” packages. Sankey diagrams and forest plots were plotted by [<https://www.bioinformatics.com.cn>] and the CNSknowall platform [<https://cnsknowall.com>] respectively. All analyses employed stepwise adjustment for confounding variables, with statistical significance defined as two-tailed P-values < 0.05.

Results

Baseline characteristics of the study population

A total of 36,718 individuals [8286 (22.6%) in CHARLS, 5225 (14.2%) in ELSA, 10,849 (29.5%) in HRS and 12,358 (33.7%) in MHAS] were included in the study population. The main baseline characteristic distributions of participants and the demographic descriptions stratified by depression status are presented in Fig. 1 and Table 1. The mean follow-up duration across the four cohorts was

9 years, with cohort-specific durations as follows: 4 years for CHARLS, 10 years for both ELSA and HRS, and 12 years for MHAS. Among the continuous variables, the mean age of participants was 63.69 (SD 9.99) years. The mean of BMI was 27.27 (SD 5.40) and the mean of WHtR was 0.55 (SD 0.10). T-tests and chi-square tests showed participants who had experienced depression tended to have a higher BMI (all $P < 0.05$) and WHtR in ELSA and HRS and lower baseline cognitive test score in all cohorts (all $P < 0.001$). In all four studies, age, gender, education status, marital status, hypertension, stroke, drinking status, and smoking status had impacts on depressive symptoms (all $P < 0.05$). Moreover, Additionalfile 1: Fig. S4 and Fig.S5 shows basic characteristics of the dynamic transition of depression-obesity status from baseline to outcome wave. Overall, the proportion of participants in all studies experiencing a state transition was relatively small.

Association between baseline depression-obesity status and changes in cognitive function

No matter which obesity indicator was adopted, compared to participants in “Neither condition,” individuals in “Depression alone” and “Comorbidity” at baseline exhibited significantly poorer performance in subsequent cognitive assessments (Fig. 2 and Additional File 1: Table S5). This association persisted even after full adjustment for all covariates, maintaining statistical significance across analytical models (“Depression alone” of Fig. 2A—CHARLS: $\beta = -0.30$, 95% CI -0.35 to -0.25 ; ELSA: $\beta = -0.27$, 95% CI -0.40 to -0.14 ; HRS: $\beta = -0.17$, 95% CI -0.26 to -0.09 ; MHAS: $\beta = -0.19$, 95% CI -0.23 to -0.14 ; Fig. 2B: CHARLS: $\beta = -0.29$, 95% CI -0.34 to -0.23 ; ELSA: $\beta = -0.24$, 95% CI -0.37 to -0.11 ; HRS: $\beta = -0.14$, 95% CI -0.23 to -0.05 ; “Comorbidity” of Fig. 2A: CHARLS: $\beta = -0.25$, 95% CI -0.42 to -0.08 ; ELSA: $\beta = -0.26$, 95% CI -0.41 to -0.11 ; HRS: $\beta = -0.18$, 95% CI -0.26 to -0.10 ; MHAS: $\beta = -0.07$, 95% CI -0.12 to -0.01 ; Fig. 2B: CHARLS: $\beta = -0.34$, 95% CI -0.45 to -0.24 ; ELSA: $\beta = -0.32$, 95% CI -0.48 to -0.16 ; HRS: $\beta = -0.26$, 95% CI -0.34 to -0.18 , all $P < 0.05$). When the obesity indicator was BMI, we observed individuals in “Obesity alone” exhibited superior cognitive development in HRS and MHAS (HRS: $\beta = 0.05$, 95% CI 0.01 to 0.08; MHAS: $\beta = 0.06$, 95% CI 0.02 to 0.10, all $P < 0.05$), whereas ELSA revealed an inverse pattern, with individuals with obesity exhibiting comparatively impaired cognitive outcomes ($\beta = -0.10$, 95% CI -0.17 to -0.02 , $P = 0.012$). When the obesity indicator was WHtR, “Obesity alone” participants demonstrated significantly poorer cognitive performance compared to “Neither condition” only in ELSA ($\beta = -0.11$, 95% CI -0.18 to -0.03 , $P = 0.004$).

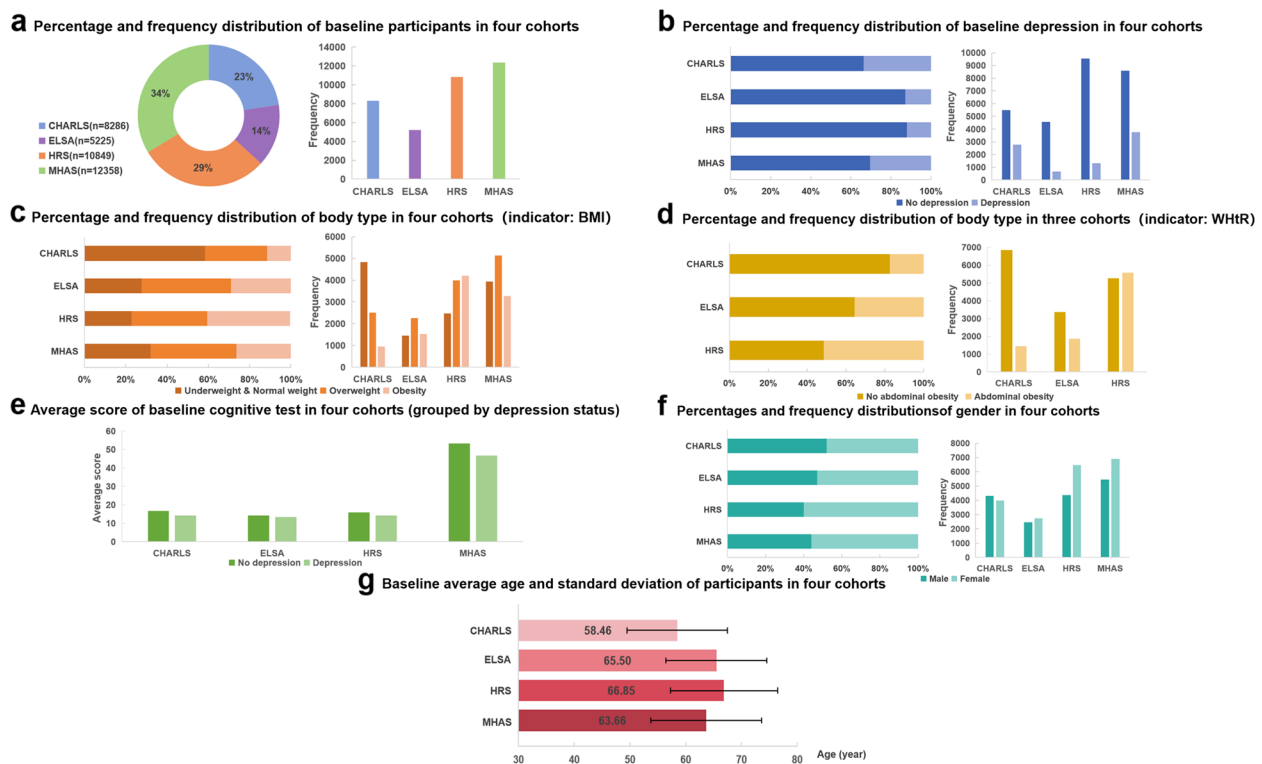


Fig. 1 Main baseline characteristics of participants from four cohorts. Abbreviations: CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; MHAS, the Mexican Health and Aging Study; HRS, Health and Retirement Study

Cluster analysis and joint trajectories of obesity and depression

In order to determine the appropriate number of potential trajectories of BMI (kg/m^2)/WHtR and depressive symptoms groups, our study combined the iterative results of Kml3D (the number of clusters was iteratively increased from 2 to 20), cluster classification metrics, and interpretability of the results [a reference group (Normal weight/No abdominal obesity & No depression) should be defined at least]. Therefore, we identified different numbers of trajectories for different cohorts respectively (Fig. 3, Additional file 1: Table S6 and S7). Furthermore, after randomly excluding 20% of participants and repeating the cluster analysis, no substantial differences were found in the newly divided trajectory group (Additional file 1: Fig. S6, Table S6 and S7).

Association between joint trajectories and changes in cognitive function

Figure 4 and Additional file 1: Table S8–S9 summarize the association between joint trajectories (BMI/WHtR and depressive symptoms) and changes in cognitive function. In Fig. 4A (adjusted all covariates), compared to the reference group (Group1: Normal weight and No depression), participants in the joint trajectory

groups with depression (regardless of obesity trajectories) had poorer cognitive function [CHARLS (Group2: Normal weight & Depression): $\beta = -0.35$, 95% CI -0.42 to -0.28 ; ELSA (Group2: Overweight & Depression): $\beta = -0.32$, 95% CI -0.44 to -0.20 ; HRS (Group2: Overweight & Depression): $\beta = -0.20$, 95% CI -0.27 to -0.13 , MHAS (Group3: Overweight & Depression): $\beta = -0.15$, 95% CI -0.19 to -0.10 ; all $P < 0.001$]. However, in other trajectory groups, inconsistent results were observed among the four studies. When compared with the reference group, the significantly accelerated cognitive decline was reported in Group 4 (Obesity and No depression) of ELSA ($\beta = -0.16$, 95% CI -0.26 to -0.06 , $P = 0.002$). Additionally, the association between trajectory groups characterized by overweight/obesity and better cognitive development was observed in HRS and MHAS [HRS (Group3 and Group4: Obesity (I)/Obesity (II) and No depression): $\beta = 0.06$, 95% CI 0.02 to 0.11 ; $\beta = 0.10$, 95% CI 0.04 to 0.16 ; MHAS (Group2: Overweight and No depression): $\beta = 0.09$, 95% CI 0.04 to 0.13 , all $P < 0.05$]; however, this association was not observed in the CHARLS.

Figure 4B shows the relationship between the joint trajectory (WHtR and depressive symptoms) and cognitive changes. In the fully adjusted model (adjusted all

Table 1 Baseline characteristics of participants from four cohorts according to depression status

Variables	China Health and Retirement Longitudinal Study (CHARLS) (n=8286)				English Longitudinal Study of Ageing (ELSA) (n=5225)				Health and Retirement Study(HRS) (n=10849)				Mexican Health and Aging Study (MHAS) (n=12358)			
	Overall (Mean ± SD or events/n)	No depression (n=5504)	Depression (n=2782)	p	No depression (n=4552)	Depression (n=673)	p	No depression (n=9539)	Depression (n=1310)	p	No depression (n=8591)	Depression(n=3767)	p			
Age (years)	63.69 ± 9.99	58.12 ± 8.97	59.13 ± 8.97	<0.001	65.39 ± 8.92	66.26 ± 9.78	0.200	66.98 ± 9.56	65.83 ± 9.97	<0.001	63.28 ± 9.77	64.53 ± 10.20	<0.001			
BMI (kg/m²)*	27.27 ± 5.40	23.82 ± 3.71	23.20 ± 3.71	<0.001	27.88 ± 4.75	28.37 ± 5.31	0.014	29.21 ± 5.71	30.52 ± 6.67	<0.001	27.81 ± 4.56	27.50 ± 4.71	0.004			
Underweight	300/728	260 (4.7)	216 (7.8)	<0.001	27 (0.6)	5 (0.7)	0.041	53 (0.6)	10 (0.8)	<0.001	88 (1.0)	69 (1.8)	<0.001			
Normal weight	2771/11579	2811 (51.1)	1544 (55.5)		1237 (27.2)	180 (26.7)		2152 (22.6)	257 (19.6)		2608 (30.4)	1180 (31.3)				
Overweight	2838/13889	1755 (31.9)	745 (26.8)		1993 (43.8)	263 (39.1)		3575 (37.5)	416 (31.8)		3728 (43.4)	1414 (37.5)				
Obesity	2233/9952	678 (12.3)	277 (10.0)		1295 (28.4)	225 (33.4)		3579 (39.4)	627 (47.9)		2167 (25.2)	1104 (29.3)				
Waist-to-height ratio (WHtR)	0.55 ± 0.10	0.54 ± 0.06	0.54 ± 0.66	0.305	0.58 ± 0.07	0.59 ± 0.08	<0.001	0.60 ± 0.09	0.63 ± 0.10	<0.001	NA	NA	NA			
No abdominal obesity	3182/15478	4557 (82.8)	2285 (82.1)	0.455	2966 (65.2)	401 (59.6)	0.005	4773 (50.0)	496 (37.9)	<0.001	NA	NA	NA			
Abdominal obesity	1583/8882	947 (17.2)	497 (17.9)		1586 (34.8)	272 (40.4)		4766 (50.0)	814 (62.1)							
Gender (%)																
Male	2955/16614	3099 (56.3)	1215 (43.7)	<0.001	2247 (49.4)	226 (33.6)	<0.001	4013 (42.1)	363 (27.7)	<0.001	4300 (50.1)	1151 (30.6)	<0.001			
Female	5577/20104	2405 (43.7)	1567 (56.3)		2305 (50.6)	447 (66.4)		5526 (57.9)	947 (72.3)		4291 (49.9)	2616 (69.4)				
Race (%)																
White	1655/14125	NA	NA	NA	4511 (99.1)	655 (97.3)	<0.001	7959 (83.4)	1000 (76.3)	<0.001	NA	NA	NA			
Other	328/1949				41 (0.9)	18 (2.7)		1580 (16.6)	310 (23.7)							
Education status (%)																
Less than lower secondary	6796/21571	4645 (84.4)	2588 (93.0)	<0.001	1678 (36.9)	333 (49.5)	<0.001	1502 (15.7)	410 (31.3)	<0.001	6950 (80.9)	3465 (92.0)	<0.001			
Upper secondary & vocational training	1395/11285	734 (13.3)	177 (6.4)		2198 (48.3)	284 (42.2)		5708 (59.8)	731 (55.8)		1250 (14.6)	203 (5.4)				
Tertiary	341/3862	125 (2.3)	17 (0.6)		676 (14.9)	56 (8.3)		2329 (24.4)	169 (12.9)		391 (4.6)	99 (2.6)				
Residence (%) [§]																
Type A(Urban)	5016/24405	2331 (42.4)	877 (31.50)	<0.001	4374 (96.1)	632 (93.9)	0.008	6381 (66.9)	912 (69.6)	0.490	6303 (73.4)	2595 (68.9)	<0.001			
Type B(Rural)	3516/12313	3173 (57.6)	1905 (68.5)		178 (3.9)	41 (6.1)		3158 (33.1)	398 (30.4)		2288 (26.6)	1172 (31.1)				
Marital status (%)																
Living alone (No)	2670/9025	462 (8.4)	419(15.1)	<0.001	1085 (23.8)	293 (43.5)	<0.001	2658 (27.9)	592 (45.2)	<0.001	2150 (25.0)	1366 (36.3)	<0.001			
Living with partner(Yes)	5862/27693	5042 (91.6)	2363 (84.9)		3467 (76.2)	380 (56.5)		6881 (72.1)	718 (54.8)		6441 (75.0)	2401 (63.7)				
Household economic situation %)*																
Low tertile	3633/12753	1736 (31.5)	1023 (36.8)	<0.001	1490 (32.7)	251 (37.3)	0.062	2933 (30.7)	684 (52.2)	<0.001	2961 (34.5)	1675 (44.5)	<0.001			
Medium tertile	2678/11751	1796 (32.6)	972 (34.9)		1530 (33.6)	214 (31.8)		3243 (34.0)	378 (28.9)		2504 (29.1)	1114 (29.6)				
High tertile	2221/12214	1972 (35.8)	787 (28.3)		1532 (33.7)	208 (30.9)		3363 (35.3)	248 (18.9)		3126 (36.4)	978 (26.0)				
Hyper-tension (%)																

Table 1 (continued)

Variables	Overall (Mean ± SD or events/n)	China Health and Retirement Longitudinal Study (CHARLS) (n=8286)			English Longitudinal Study of Ageing (ELSA) (n=5225)			Health and Retirement Study(HRS) (n= 10849)			Mexican Health and Aging Study (MHAS) (n=12358)		
		No depression (n=5504)	Depression (n=2782)	p	No depression (n=4552)	Depression (n=673)	p	No depression (n=9539)	Depression (n=1310)	p	No depression (n=8591)	Depression(n=3767)	p
No	4317/20672	4148 (75.4)	2014 (72.4)	0.003	2791 (61.3)	366 (54.4)	<0.001	4723 (49.5)	544 (41.5)	<0.001	4693 (54.6)	1393 (37.0)	<0.001
	Yes	4215/16046	768 (27.6)		1761 (38.7)	307 (45.6)		4816 (50.5)	766 (58.5)		3898 (45.4)	2374 (63.0)	
Diabetes (%)													
No	6758/31007	5179 (94.1)	2587 (93.0)	0.050	4233 (93.0)	601 (89.3)	<0.001	8076 (84.7)	955 (76.0)	<0.001	6761 (78.7)	2615 (69.4)	<0.001
Yes	1734/5671	325 (5.9)	195 (7.0)		319 (7.0)	72 (10.7)		1463 (15.3)	315 (24.0)		1830 (21.3)	1152 (30.3)	
Cancer (%)													
No	8147/34546	5457 (99.1)	2756 (99.1)	0.711	4239 (93.1)	619 (92.0)	0.277	8331 (87.3)	1169 (89.2)	0.051	8372 (97.5)	3603 (95.6)	<0.001
Yes	385/2172	47 (0.9)	26 (0.9)		313 (6.9)	54 (8.0)		1208 (12.7)	141 (10.8)		219 (2.5)	164 (4.4)	
Stroke (%)													
No	8137/35451	5422 (98.5)	2682 (96.4)	<0.001	4407 (96.8)	637 (94.7)	0.004	9074 (95.1)	1218 (93.0)	0.001	8411 (97.9)	3600 (95.6)	<0.001
Yes	395/1267	82 (1.5)	100 (3.6)		145 (3.2)	36 (5.3)		465 (4.9)	92 (7.0)		180 (2.1)	167 (4.4)	
Arthritis (%)													
No	4800/23580	4078 (74.1)	1557 (56.0)	<0.001	3148 (69.2)	328 (48.7)	<0.001	4515 (47.3)	418 (31.9)	<0.001	7039 (81.9)	2497 (66.3)	<0.001
Yes	3732/13138	1426 (25.9)	1225 (44)		1404 (30.8)	345 (51.3)		5024 (52.7)	892 (68.1)		1552 (18.1)	1270 (33.7)	
Drinking status (%)													
None	5660/19578	3134 (56.9)	1713 (61.6)	<0.001	389 (8.5)	117 (17.4)	<0.001	4187 (43.9)	741 (56.6)	<0.001	6208 (72.3)	3089 (82.0)	<0.001
Yes	2872/17140	2370 (43.1)	1069 (38.4)		4163 (91.5)	556 (82.6)		5352 (56.1)	569 (43.4)		2383 (27.7)	678 (18.0)	
Smoking status (%)													
None	4908/17718	3042 (55.3)	1698 (61.0)	<0.001	1727 (37.9)	206 (30.6)	<0.001	4249 (44.5)	537 (41.0)	0.015	5136 (59.8)	2467 (65.5)	<0.001
Yes	3624/18740	2462 (44.7)	1084 (39.0)		2825 (62.1)	467 (69.4)		5290 (55.5)	773 (59.0)		3455 (40.2)	1300 (34.5)	
Baseline cognitive test score	NA	16.65 ± 4.63	14.33 ± 4.63	<0.001	14.25 ± 3.40	13.42 ± 3.64	<0.001	15.92 ± 3.98	14.29 ± 4.60	<0.001	53.22 ± 18.25	46.71 ± 17.71	<0.001

Data are presented as means ± standard deviations or number (proportion %). P values were calculated using analysis of t-test and Cchi-square test for continuous and categorical variables, respectively. BMI: In CHARLS: underweight (< 18.5 kg/cm²); normal weight (>= 18.5 kg/cm² and < 22.9 kg/cm²); overweight (>= 23 kg/cm² and < 28 kg/cm²); obesity (>= 28 kg/cm²). In other databases, underweight (< 18.5 kg/cm²); normal weight (>= 18.5 kg/cm² and < 25 kg/cm²); overweight (>= 25 kg/cm² and < 30 kg/cm²); obesity (>= 30 kg/cm²). Residence: In CHARLS, HRS and MHAS: type A: Urban area; type B: Rural area. In ELSA, type A: UK; type B: Other area. Household economic situation: In CHARLS, ELSA and MHAS, Household economic situation was defined by total household per capita consumption. In HRS, Household economic situation was defined by total household income

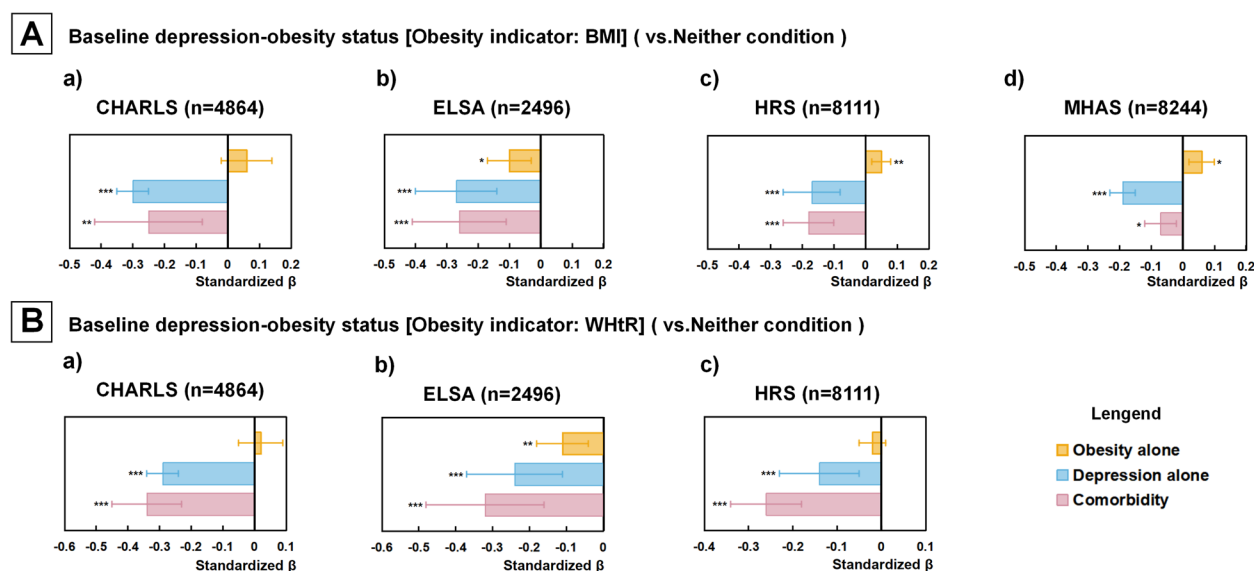


Fig. 2 Associations between joint trajectories and changes in cognitive function **A** Association between baseline depression-obesity status (Obesity indicator: BMI) and changes in cognitive function. **B** Association between baseline depression-obesity status (Obesity indicator: WHtR) and changes in cognitive function. All models were adjusted for age, gender, education status, residence, marital status, household economic situation, (race:only in the ELSA, and HRS), hypertension, diabetes, cancer, stroke, arthritis, drinking status, and smoking status. [Standardized β are shown as centers of horizontal bar plots, with error bars representing 95% CI, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$]. Abbreviations: CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; MHAS, the Mexican Health and Aging Study; HRS, Health and Retirement Study. Notes: In CHARLS, ELSA, and MHAS, household economic situation was defined by total household per capita consumption. In HRS, household economic situation was defined by total household income

covariates), compared to the reference group (Group1: No abdominal obesity & No depression), participants in “Abdominal obesity and Depression group” met with more cognitive decline [ELSA (Group2: Abdominal obesity (I) and Depression): $\beta = -0.20$, 95% CI 0.29 to -0.12 ; HRS (Group2: Abdominal obesity (I) and Depression): $\beta = -0.20$, 95% CI: -0.27 to -0.14 , all $P < 0.001$]. In CHARLS, although no “Abdominal obesity and Depression” group was divided, the group containing depression (Group 2: No abdominal obesity (II) and Depression) also had poorer cognitive development than the reference group ($\beta = -0.31$, 95% CI -0.39 to -0.23 , $P < 0.001$); meanwhile, participants in this group exhibited a higher WHtR compared to those in the reference group (Additional file 1: Table S9).

Subgroup analysis

We further stratified the population into different subgroups according to main demographic variables (such as age, gender, education status and so on) to assess the heterogeneity of joint trajectories (BMI and Depressive symptoms and WHtR and Depressive symptoms) on changes in cognitive function (Additional file 2: Fig. S7–S8, and Table S10). The results were similar to the main results but not consistent in four cohorts.

Secondary and sensitivity analyses

The current research further conducted two secondary analyses to supplement the results. (1) We explored the association between the transition in depression-obesity status and development of cognitive function in participants. The results indicated that compared to those who maintained neither depression nor obesity, participants in other transition groups may have had worse cognitive function (Additional file 3: Table S11–S13 and Fig. S9). (2) We included Weight-Adjusted Waist Index (WWI) as another measure of obesity. It was observed that individuals in the joint trajectory groups with higher WWI and/or depression had poorer cognitive development compared to the reference group (Low WWI and No depression) (Additional file 3: Fig. S10 and Table S14–S15). We also conducted a sensitivity analysis (removed 566 underweight participants) and obtained results that accorded closely with those of the main analyses (Additional file 3: Fig. S11 and Table S16).

Discussion

Our analysis of longitudinal data from four multinational cohorts examined the additive and synergistic effects of depression and obesity/abdominal obesity (AO), and their combination, on cognitive function in middle-aged and older adults. Specifically, individuals in “Depression

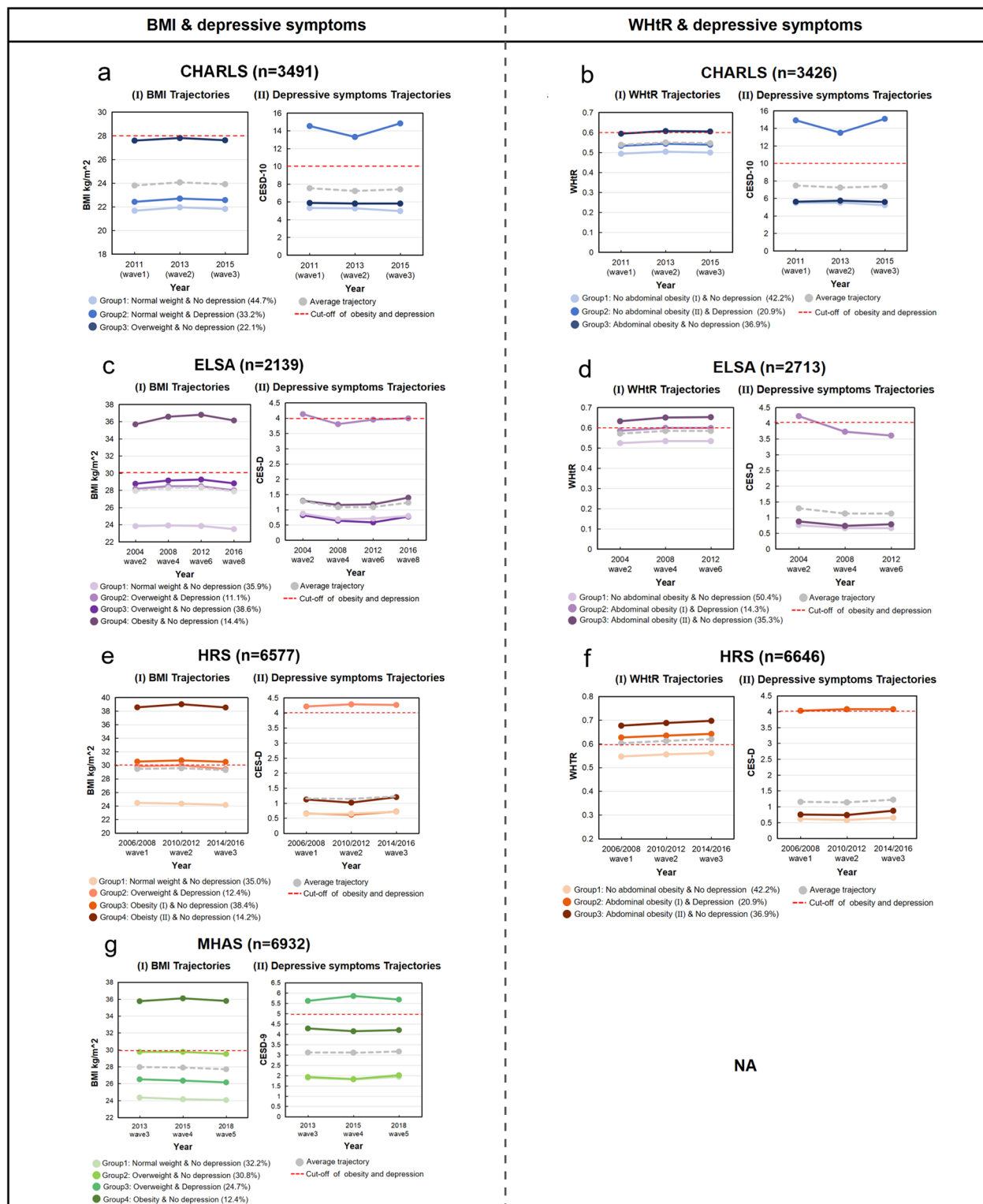


Fig. 3 Joint trajectories of BMI (kg/m²)/WHtR and depressive symptoms in all waves. Abbreviations: CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; MHAS, the Mexican Health and Aging Study; HRS, Health and Retirement Study; 95% CI, 95% confidence interval. Notes: The different colored lines show the mean of BMI/WHtR and score of depressive symptoms in different trajectory groups; (I) and (II) represent different degrees

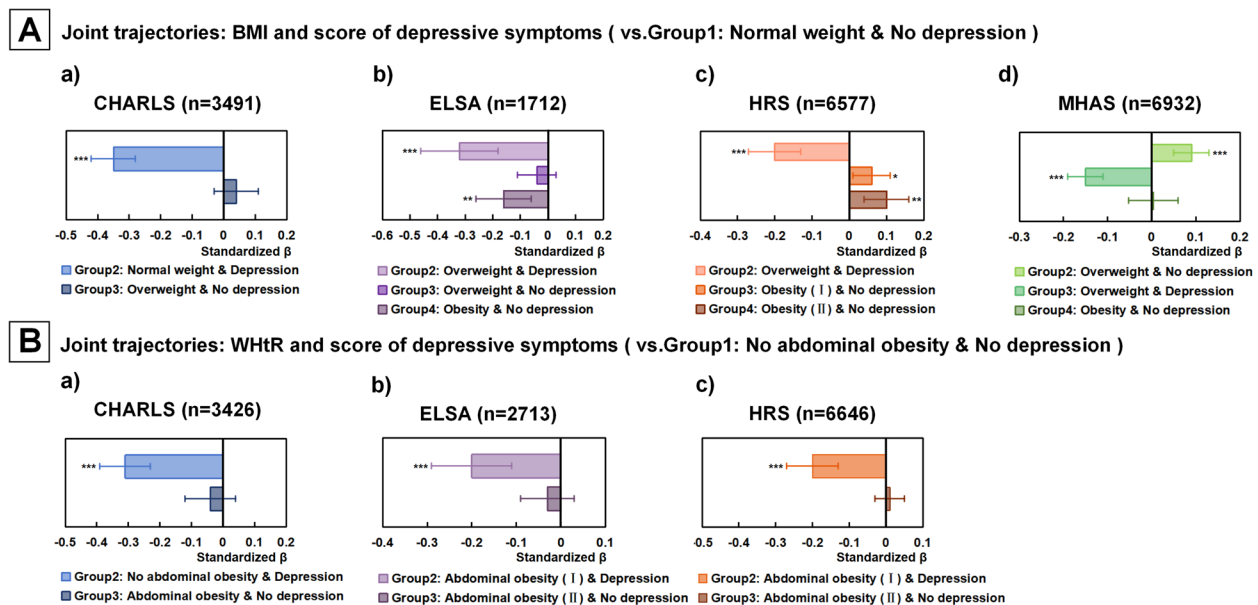


Fig. 4 Associations between joint trajectories and changes in cognitive function. **A** Associations between joint trajectories (BMI and score of depressive symptoms) and changes in cognitive function. **B** Associations between joint trajectories (WHtR and score of depressive symptoms) and changes in cognitive function. All models were adjusted for age, gender, education status, residence, marital status, household economic situation, race (race:only in the ELSA and HRS), hypertension, diabetes, cancer, stroke, arthritis, drinking status, and smoking status [Standardized β are shown as centers of horizontal bar plots, with error bars representing 95% CI, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$]. Abbreviations: CHARLS, China Health and Retirement Longitudinal Study; ELSA, English Longitudinal Study of Ageing; MHAS, the Mexican Health and Aging Study; HRS, Health and Retirement Study. Notes: In CHARLS, ELSA, and MHAS, household economic situation was defined by total household per capita consumption. In HRS, household economic situation was defined by total household income. (I) and (II) represented different degrees

alone” or “Comorbidity” at baseline experienced accelerated cognitive decline compared to those with “Neither condition,” using either BMI or WHtR. We also found the impact of “Obesity alone” was heterogeneous. However, when WHtR defined obesity, “Comorbidity” showed the greatest variance to variance in cognitive development. In cluster analysis, employing BMI to gauge obesity revealed that the joint trajectory of depression and obesity had heterogeneous effects on cognition decline. Nonetheless, all cohorts demonstrated that the trajectory group with depression had quicker cognitive deterioration than the reference group. Notably, in examining AO, a key obesity subtype, all three groups with measured waist circumference revealed that the “Higher WHtR/AO & Depression” group significantly contributed to cognitive decline relative to the reference group. The outcome was further corroborated through the joint trajectory analysis of secondary analysis employing WWI.

Factors influencing the cognitive function of middle-aged and older adults have conducted numerous cross-cultural longitudinal studies on the relationship between obesity and cognition, as well as depression and cognition individually [9, 19, 20, 27, 28, 31]. However, considering that obesity and depression may share common pathological pathways [8], and serve as significant factors

affecting overall systemic metabolic disorders and quality of life [69–74], exerting negative influences on higher-order functions of the brain. Therefore, results evaluating only individual factors may be limited and biased, which makes them difficult to interpret and validate [73].

Replicated evidence indicates a negative relationship exists between cognitive function and depressive disorders [11, 75–78]. Cognitive decline in middle- and older-aged adults with depression may be a consequence of alteration in the structure and function of pathogenetic mechanisms including neurotransmitters, vascular processes, neuroinflammation, and neural networks [79–83].

The association between obesity and cognition is inconsistent and heterogeneity [26, 27, 84]. Some studies have indicated a negative correlation between obesity and cognitive ability [27, 84]. In contrast, other research has shown a positive correlation, implying that obesity may exert a protective effect on cognitive development in middle-aged and older adults. When employing BMI as an indicator of obesity and comprehensively considering the depressive symptoms of the participants, our study also reached similar conclusions among the HRS and MHAS populations. This could be explained by the “obesity paradox,” which suggests that compared to underweight individuals, body fat may have a protective

effect on specific health conditions, including cognitive function [85]. This can be explained as higher body fat may protect against frailty, or that people with obesity may have higher energy reserves to buffer the effects of age-related brain atrophy [26, 27]. Additionally, individuals with overweight or obesity may also attempt to lose weight by engaging in physical exercise, and appropriate physical activity has been shown to improve cognitive conditions [86, 87]. For underweight individuals, this association may be linked to sarcopenia and malnutrition [88]. Meanwhile, weight loss or localized adipose tissue deficiency is due to reduced food intake and food attractiveness [89]. Furthermore, recent research has highlighted that the obesity paradox may be explained by the interaction between obesity and other factors, such as individual genetic, metabolic, and anthropometric factors [90–92].

Agreement exists that BMI is an imprecise estimate of fat mass which may be a source of variance in the reported studies [93]. Differences in body fat distribution exhibit different patterns in relation to cognitive function [94]. As a general measure of body weight, BMI fails to reflect fat distribution. In contrast, the Waist-to-Height Ratio (WHtR) captures central obesity and serves as a surrogate marker for visceral fat accumulation. Visceral fat is closely associated with insulin resistance, endothelial dysfunction, and elevated levels of pro-inflammatory cytokines such as interleukin-6 (IL-6) and C-reactive protein (CRP), all of which are considered key factors in the pathogenesis of cognitive impairment and neurodegenerative diseases [95]. Additionally, the Weight-Adjusted Waist Index (WWI) provides an adiposity index independent of total body weight. Studies have shown that WWI correlates more strongly with cardiovascular metabolic risk factors, oxidative stress, and inflammatory markers than BMI. Therefore, it is imperative to further investigate the relationship between obesity subtypes or fat distribution and cognitive function. Based on this, the present study also conducted a parallel analysis using the WHtR as a measure of central obesity and obtained relatively consistent results across multinational samples. Previous research indicates that AO is a risk factor for cognitive decline in later life [96]. This association may be related to dietary habits, metabolic impairment, lipid disorders and activities, etc. [97–99]. However, the decline in the physical and mental functions of older adults leads to lower levels of leisure activity [100], which may promote the occurrence of AO and further impair their cognitive functions, while there may be a mutually reinforcing relationship between cognitive decline and some of the aforementioned influencing factors. Therefore, the relationship between fat distribution/physique and cognition requires further consideration of more

specific fields and possibly integrate multiple physiological indicators for exploration. This point has been partially reflected in this study.

Overall, obesity and depression are two prevalent and potentially interdependent metabolic disorders, imposing a significant disease burden on the health and well-being of the middle-aged and older population. This study comprehensively examined the impact of the joint trajectories of these two diseases on the cognitive development of the research subjects and, to some extent, revealed the intricate connections between them and cognitive decline. The comorbidity of obesity (obesity indicator: BMI) and depression at baseline may have a synergistic negative effect on cognitive function. When including the scale of time, the conclusions drawn from different cohorts were inconsistent. These differences may stem from the influence of cultural differences, dietary habits, and the geographical environment [101–103]. For instance, a Mediterranean diet, rich in antioxidants and healthy fats, has been associated with better cognitive function even in people living with obesity, whereas high-fat, high-sugar diets common in Western countries may exacerbate cognitive decline [101]. Conversely, after conducting a parallel analysis employing WHtR as an obesity indicator to reflect the accumulation of abdominal fat, we have identified its potential negative impact on cognitive function in combination with depression. This may be due to two shared pathways.

Firstly, both conditions elevate the body's inflammatory levels. Depressed individuals have higher levels of inflammatory markers in their blood, such as interleukin (IL)-1 β and C-reactive protein (CRP) [104, 105]. Concurrently, adipose tissue has been identified as a major source of inflammatory molecules [106], and the link between inflammatory cytokines and AO is closer than that with BMI [107], suggesting that the accumulation of abdominal fat may predispose the body to a pro-inflammatory state [108]. It is worth noting that circulating pro-inflammatory mediators are associated with neurodegenerative diseases, and peripheral inflammation has been shown to negatively impact the development of chronic brain inflammation [109]. These insights imply that depression and AO may jointly contribute to an increase in systemic inflammation, which could subsequently impair cognitive function in middle-aged and older adults. Additionally, the combined effects of AO and depression are manifested in the elevation of oxidative stress levels. Secondly, during the aging process, the brain's antioxidant defense system becomes increasingly imbalanced, leading to increased oxidative stress, which is one of the primary causes of cognitive decline [110]. It is widely accepted that abdominal fat is a serious risk factor for the accumulation of visceral fat, which

promotes the release of pro-oxidants, pro-inflammatory agents, and reactive oxygen species (ROS) [111, 112]. Studies have also demonstrated that antioxidant defenses are weakened in individuals with depression, thereby increasing oxidative stress [113]. In summary, AO and depression may jointly elevate oxidative stress levels within the body, which can stimulate brain atrophy and morphological changes within the brain that ultimately result in cognitive decline [114].

This study has several strengths. Firstly, analyzed multinational cohorts and a large sample size improve the generalizability of the results. Secondly, advanced and appropriate analytical methods provide a more comprehensive interpretation of the joint contributions of the heterogeneous trajectories of obesity and depression to cognitive function. Additionally, the current study examined multiple obesity-related indicators, which will more accurately assess the interrelationship between obesity and health status comprehensively. Finally, we confirmed the robustness of our findings through various sensitivity analyses. To sum up, our findings underscore the importance of early detection and management of both depression and obesity as a potential strategy to mitigate cognitive deterioration. Specifically, clinicians should screen for depressive symptoms alongside obesity assessments (e.g., using BMI or waist-to-height ratio) to identify at-risk individuals early, allowing for timely intervention. Moreover, an integrated treatment approach can be adopted for patients with obesity-depression comorbidity. Combining pharmacotherapy for depression with lifestyle modifications aimed at weight management and improved physical activity can enhance both mental and physical well-being, thereby potentially slowing cognitive decline.

Of note, there are several limitations to be acknowledged. Firstly, there is a potential self-reporting bias. Height and weight data in the MHAS cohort were self-reported, potentially misclassifying obesity prevalence and obscuring its relationship with cognitive decline, as BMI accuracy remains unverified. Secondly, there are variations in measurement dimensions of cognitive function across different cohorts. Although the study attempted to standardize scores from each cohort by generating z-scores, it must be acknowledged that disparities in cognitive measurement instruments and their respective sensitivities may introduce a degree of systematic bias, potentially affecting outcome validity and generalizability. Thirdly, there are unmeasured confounding factors. Although the study adjusted for multiple confounding factors, certain variables relevant to cognitive function—such as dietary patterns [101], physical activity levels [86], social support networks [115], and sleep quality [116]—were not consistently available across the

included cohorts due to limitations in data collection or heterogeneity in measurement methods. The absence of these factors may have influenced the observed associations. These constraints suggest that the findings should be generalized with caution to populations with differing sociodemographic characteristics, measurement infrastructures, or exposure to unmeasured confounders—particularly among groups who may experience difficulty providing accurate and complete data. And to enhance generalizability, future research must prioritize: [1] replication in geographically and culturally diverse cohorts, [2] integration of objectively measured variables (e.g., accelerometer-derived physical activity, validated dietary assessments), and [3] systematic inclusion of previously unmeasured confounders. Such efforts are essential to validate the observed associations across heterogeneous populations and real-world clinical contexts.

Conclusions

Herein, we observed that the comorbidity of obesity and depression independently and additively negatively affected cognitive function in middle-aged and older people. This advocates for clinical screening to prioritize depression and abdominal obesity beyond BMI, thereby enhancing early identification of high-risk populations. Overall, the critical clinical importance of managing abdominal fat accumulation and maintaining positive emotions to prevent and mitigate cognitive deterioration is emphasized. For comorbid cases, an innovative dual-pathway intervention could be considered: combining antidepressant pharmacotherapy with lifestyle-based weight management to simultaneously improve mental/physical health and preserve cognition.

Abbreviations

AO	Abdominal obesity
BMI	Body Mass Index
CCCE	Cross-Cultural Cognitive Examination
CESD	Center for Epidemiologic Studies Depression Scale
CESD-9	9-item version of the CESD (used in MHAS)
CESD-10	10-item version of the CESD (used in CHARLS)
CHARLS	China Health and Retirement Longitudinal Study
CI	Confidence interval
CRP	C-reactive protein
ELSA	English Longitudinal Study of Ageing
GEE	Generalized Estimating Equations
GLM	Generalized linear model
HPA	Hypothalamic-pituitary-adrenal (axis)
HRS	Health and Retirement Study
IL-6	Interleukin-6
KMI3D	K-means longitudinal clustering in 3 dimensions
MHAS	Mexican Health and Aging Study
MMSE	Mini-Mental State Examination
ROS	Reactive oxygen species
SD	Standard deviation
TICS	Telephone Interview for Cognitive Status
WtHr	Waist-to-height ratio
WHO	World Health Organization
WWI	Weight-Adjusted Waist Index
z-score	Standardized score

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12916-025-04298-2>.

Additional File 1

Acknowledgements

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Authors' contributions

RW, YC and BC contributed to the study design. RW analysed the data and RW, YC and BC were involved in data interpretation. RW and YC drafted the first version of the manuscript and prepared tables and figures. KMT, RSM, and BC reviewed and revised the manuscript. All authors were involved in the writing and critical revision of the manuscript. All authors have read and approved the final manuscript.

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Data availability

The datasets analyzed during the current study are publicly available and can be accessed as follows: CHARLS (China Health and Retirement Longitudinal Study): <https://hrs.isr.umich.edu/about?> ELSA (English Longitudinal Study of Ageing): <https://ukdataservice.ac.uk/about/HRS> (Health and Retirement Study): <https://charls.pku.edu.cn/gy/gyxm.htm> MHAS (Mexican Health and Aging Study): <https://www.mhasweb.org/Home/Index.aspx> All data are free to access for bona fide research purposes upon registration with the respective study platforms. No personally identifiable information was used. Analysis scripts and processed data files can be requested from the corresponding author.

Declarations

Ethics approval and consent to participate

The China Health and Retirement Longitudinal Study (CHARLS) received approval from the Ethical Review Committee of Peking University (IRB00001052-11015). The English Longitudinal Study of Ageing (ELSA) was approved by the London Multicentre Research Ethics Committee (MREC/01/2/91). The Health and Retirement Study (HRS) was approved by the Institutional Review Board at the University of Michigan and the National Institute on Aging (HUM00061128) and the Mexican Health and Aging Study (MHAS) is partly sponsored by the National Institutes of Health/National Institute on Aging (NIH R01AG018016) in the United States and the Instituto Nacional de Estadística y Geografía (INEGI) in Mexico. All respondents provided written informed consent.

Consent for publication

Not applicable.

Competing interests

Dr. Roger S. McIntyre has received research grant support from CIHR/GACD/ National Natural Science Foundation of China (NSFC) and the Milken Institute; speaker/consultation fees from Lundbeck, Janssen, Alkermes, Neumora Therapeutics, Boehringer Ingelheim, Sage, Biogen, Mitsubishi Tanabe, Purdue, Pfizer, Otsuka, Takeda, Neurocrine, Neurawell, Sunovion, Bausch Health, Axsome, Novo Nordisk, Kris, Sanofi, Eisai, Intra-Cellular, NewBridge Pharmaceuticals, Viatrix, Abbvie and Atai Life Sciences. Kayla M. Teopiz has received fees from Braxia Scientific Corp. All other authors have no conflicts of interest to declare.

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